Research Report

Forecasting Time-Varying Intermarket Dependencies Between Cryptocurrencies and Conventional Assets Using Machine Learning

Master's Thesis — Summary Report

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Contents

1 Introduction and Motivation **3**

2 Data **3**

2.1 Asset universe 3

2.2 Sample period and preprocessing 4

2.3 Dependency targets 4

3 Methodology **6**

3.1 Feature engineering 6

3.2 Forecasting models 6

3.3 Walk-forward evaluation 7

3.4 Evaluation metrics 7

3.5 Investor signal layer 7

4 Results **8**

4.1 Average forecasting performance 8

4.2 Performance across window lengths 8

4.3 Representative forecast visualisation 9

4.4 Diebold–Mariano tests 10

4.5 Temporal error dynamics 11

4.6 Investor signal layer 11

5 Key Findings and Discussion **12**
5.1 The persistence result is the main finding 12
5.2 Where machine learning adds genuine value 13
5.3 Practical interpretation of the signal layer 13

6 Limitations and Future Work **13**
6.1 Limitations 13
6.2 Directions for future research 13

7 Conclusion **14**

1 Introduction and Motivation

Financial markets are increasingly interconnected, and the relationship between cryptocurrency and conventional assets has become one of the most actively discussed topics in modern quantitative finance. Unlike traditional assets, Bitcoin and Ethereum trade continuously, respond rapidly to sentiment shifts, and can simultaneously behave as a speculative instrument, an inflation hedge, or a macro-risk proxy. This hybrid nature makes the *dependency* between crypto and conventional markets time-varying and difficult to model with standard static tools.

The central question of this thesis is:

> *Can information derived from cryptocurrency dynamics be used to forecast time-varying dependency structures with conventional assets, and can such forecasts be translated into a practically useful early-warning signal for investors?*

Answering this question requires a disciplined empirical framework that avoids the most common pitfalls in financial ML research: in-sample overfitting, look-ahead bias, and cherry-picked metrics. The thesis therefore adopts a strict walk-forward out-of-sample evaluation throughout and compares machine-learning models against both a classical econometric benchmark (DCC-GARCH) and simple persistence baselines.

The contribution of the thesis is twofold. First, it provides a **reproducible forecasting pipeline** for rolling intermarket dependencies. Second, it introduces an **investor signal layer** that converts dependency forecasts into a stress-warning signal for conventional markets.

2 Data

2.1 Asset universe

Daily market data were obtained via the `yfinance` interface. Bitcoin (`BTC-USD`) is the base cryptocurrency throughout. The conventional asset universe consists of six instruments selected to represent broad equity risk, precious metals, and macro-defensive positioning.

Table 1: Asset universe.

Ticker	Description	Category
`BTC-USD`	Bitcoin	Cryptocurrency (base)
`ETH-USD`	Ethereum	Cryptocurrency (reference)
`^GSPC`	S&P 500 index	Equity
`^IXIC`	NASDAQ Composite index	Equity
`GLD`	Gold ETF (SPDR)	Precious metals
`SLV`	Silver ETF (iShares)	Precious metals
`UUP`	U.S. Dollar Index ETF proxy	Macro / FX

2.2 Sample period and preprocessing

The sample spans **9 November 2017 to 27 February 2026**, yielding 2,976 daily observations per asset. The effective start date is determined by the earliest availability of ETH-USD on Yahoo Finance. All price series are converted to daily log returns. Missing observations due to non-overlapping trading calendars are handled by alignment and forward-filling of up to two consecutive non-trading days.

The full dataset therefore contains $7 \times 2{,}975 = 20{,}825$ daily log-return observations, with no remaining missing values after preprocessing.

2.3 Dependency targets

For each of the five Bitcoin–other-asset pairs, a rolling Pearson correlation is computed over four window lengths $w \in \{14, 30, 60, 90\}$ trading days, producing $5 \times 4 = 20$ dependency series in total.

To stabilize the distribution and reduce boundary effects, each correlation series is passed through the Fisher z-transform:

$$z_t^{(w)} = \frac{1}{2} \log\left(\frac{1 + \rho_t^{(w)}}{1 - \rho_t^{(w)}}\right),$$

where $\rho_t^{(w)}$ is the rolling correlation at time t with window w. All forecasting models use the Fisher-transformed series as the prediction target. The one-step-ahead value $z_{t+1}^{(w)}$ is the quantity to be forecast from information available at time t.

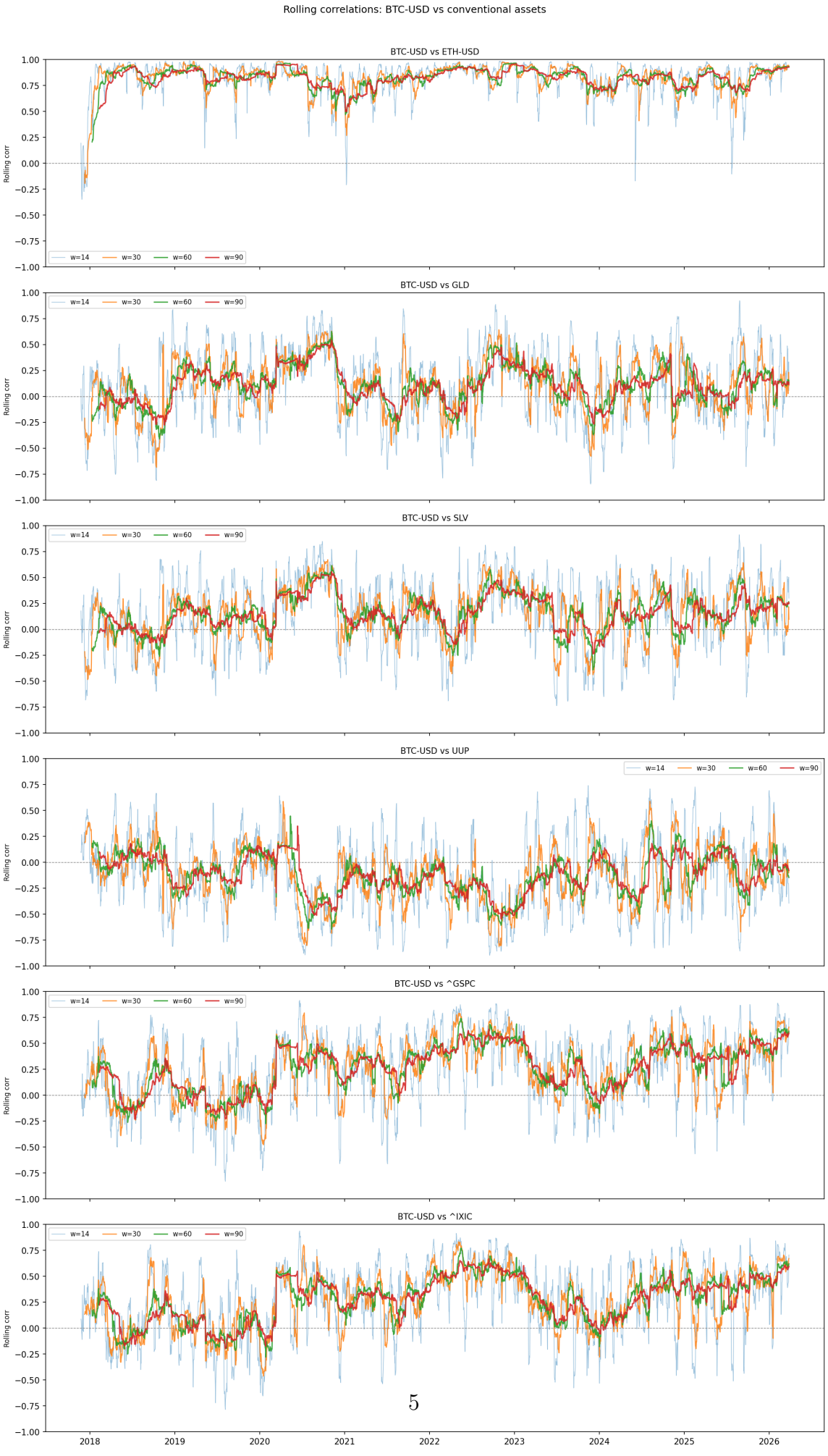


Figure 1: Rolling correlations between Bitcoin and each of the five conventional assets

3 Methodology

3.1 Feature engineering

Predictors are constructed exclusively from information available at time t or earlier, ensuring no look-ahead contamination. The feature set covers six interpretable groups:

Table 2: Feature groups used in the dependency-forecasting layer.

Group	Description
Dependency lags	Recent values of the rolling correlation target
Return lags	Lagged log-returns of BTC and the paired asset
Rolling means	Local average return over short sub-windows
Rolling volatility	Local standard deviation of returns
Spread features	Signed / absolute gap between BTC and asset behaviour
Cross-asset context	Additional lags capturing joint dynamics

Dependency lags are by far the most important predictors because the rolling-correlation target is smooth and strongly autocorrelated by construction. All features are computed inside the training window only; the test-period window never contributes to feature calculations.

3.2 Forecasting models

Eight forecasting approaches are evaluated for each of the 20 dependency experiments, covering the full spectrum from simple baselines to complex ensemble methods and a classical econometric benchmark.

Table 3: Models evaluated in the dependency-forecasting layer.

Model	Type	Notes
`Naive_Last`	Persistence baseline	$\hat{z}_{t+1} = z_t$
`AR1`	Autoregressive	OLS, one lag
`Ridge`	Regularised linear	L2 penalty, standardised features
`ElasticNet`	Regularised linear	L1+L2 penalty
`RF`	Tree ensemble	Random Forest
`GBM`	Tree ensemble	Gradient Boosting
`XGB_GPU`	Tree ensemble	XGBoost (GPU / CPU fallback)
`DCC_GARCH`	Econometric benchmark	Walk-forward DCC-GARCH(1,1)

3.3 Walk-forward evaluation

All experiments use an **expanding-window walk-forward design**. The initial training window contains the first T_0 observations (approximately 60% of the sample). At each forecast origin $t \geq T_0$:

1. The model is re-fitted on all observations up to t.
2. A forecast is produced for $t + 1$.
3. The window expands by one day.

This design mirrors real-world forecasting conditions and ensures that no future information is used in model fitting. Refitting is performed periodically (every 30 steps) to balance computational cost and temporal accuracy.

The DCC-GARCH benchmark undergoes the same expanding-window procedure, making the comparison methodologically fair. This *leakage-safe* DCC implementation is a key feature of the thesis: many studies compare ML models against a full-sample DCC fit, which inadvertently gives the benchmark access to future data.

3.4 Evaluation metrics

Forecasting accuracy is measured by:

- **RMSE** — root mean squared error (primary ranking metric);
- **MAE** — mean absolute error;
- $\boldsymbol{R^2}$ — out-of-sample coefficient of determination.

Statistical significance of model differences is assessed with the **Diebold–Mariano (DM) test** using a Newey–West long-run variance estimator. Two comparisons are made for each experiment: ML vs. DCC-GARCH and ML vs. Naive persistence.

3.5 Investor signal layer

A second modelling layer reformulates the problem as binary classification. The target is a *stress-day indicator* for the conventional asset: a day is labelled as a stress day if the asset's log return falls below a configurable negative-sigma threshold. Three classifiers are tested: **Logistic Regression**, **Random Forest (Cls)**, and **GBM (Cls)**. Signal performance is measured by Balanced Accuracy, F1 score for the downside class (F1_{down}), AUC, and exit rate.

4 Results

4.1 Average forecasting performance

Table 4 presents average out-of-sample performance across all 20 dependency experiments (5 pairs × 4 windows). Models are ranked by average RMSE.

Table 4: Average out-of-sample performance across all 20 dependency experiments.

Model	Avg. RMSE	Avg. MAE	Avg. R^2
AR1	**0.0667**	**0.0405**	**0.9421**
Naive_Last	0.0674	0.0396	0.9409
Ridge	0.0916	0.0637	0.8914
ElasticNet	0.0923	0.0644	0.8898
RF	0.1013	0.0711	0.8687
GBM	0.1038	0.0734	0.8622
XGB_GPU	0.1057	0.0758	0.8566
DCC_GARCH	0.2345	0.1848	0.2788

Three observations stand out immediately.

First, the best model overall is `AR1` (RMSE = 0.0667, $R^2 = 0.9421$). The naive persistence baseline `Naive_Last` is nearly identical, trailing by only 0.0007 RMSE. This is not a failure of the framework — it is an empirical finding about the nature of the target. Rolling correlation is a smooth, highly autocorrelated series; the primary source of one-step-ahead predictability is therefore the target's own recent history.

Second, among the ML models, regularised linear methods (Ridge and ElasticNet) outperform tree ensembles. Ridge achieves the best ML RMSE (0.0916), while XGBoost with GPU acceleration (0.1057) ranks last among ML models. This pattern is consistent with the high persistence of the target: ridge shrinkage effectively captures the dominant autoregressive structure, while boosted trees tend to overfit minor noise.

Third, the DCC-GARCH benchmark is substantially weaker than all other approaches (RMSE = 0.2345, $R^2 = 0.2788$). This gap is large and consistent across experiments.

4.2 Performance across window lengths

Table 5 shows the best-model RMSE for each window length, averaged across the five asset pairs.

Table 5: Best-model RMSE by rolling window length (average across 5 pairs). All best models are AR1.

Window w	Best RMSE (AR1)
14 days	0.1451
30 days	0.0658
60 days	0.0332
90 days	0.0228

Longer windows produce smoother, more predictable targets; RMSE drops monotonically from 0.1451 at $w = 14$ to 0.0228 at $w = 90$. This confirms that the forecasting difficulty is largely governed by the smoothness of the target rather than by the choice of model.

4.3 Representative forecast visualisation

Figure 2 illustrates the walk-forward forecast for the Bitcoin–S&P 500 pair at $w = 30$, the experiment of highest practical interest.

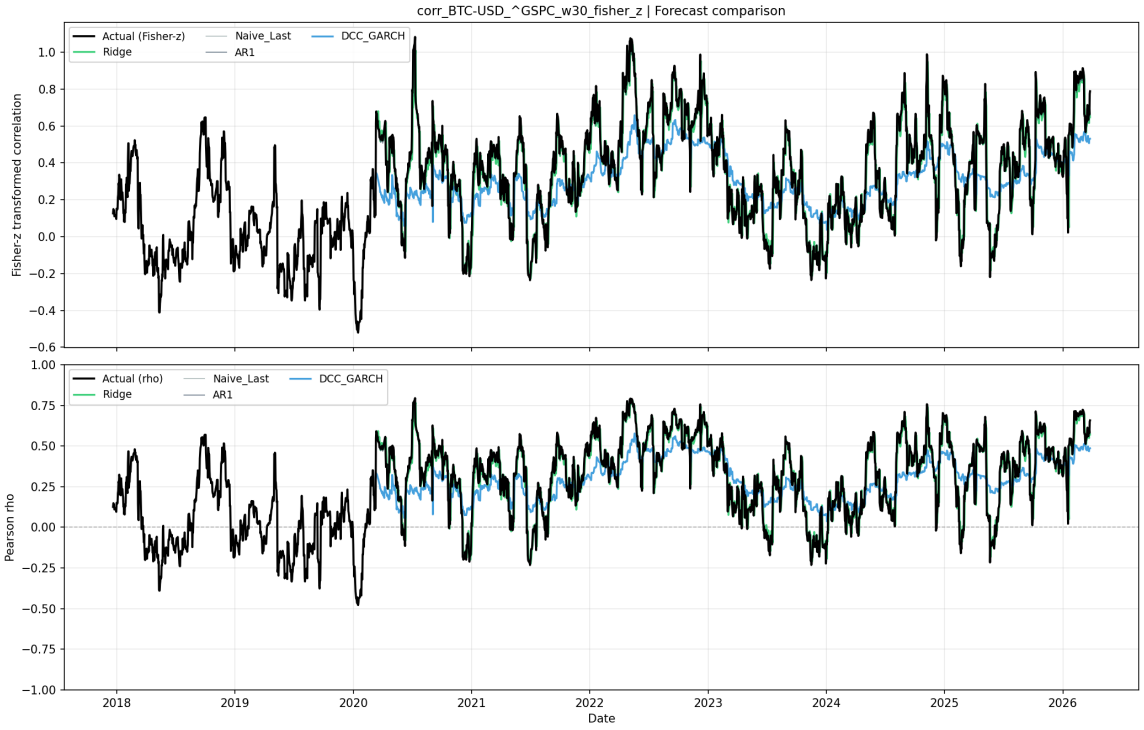


Figure 2: Out-of-sample walk-forward forecast of the Bitcoin–S&P 500 rolling dependency (Fisher-transformed, $w = 30$). The best models closely track the target, especially in low-volatility regimes.

The figure highlights a characteristic pattern: models track broad directional trends well (low-frequency evolution), but sharp regime transitions — for example, the abrupt

dependency spike during the COVID-19 market crash in March 2020 — remain difficult for all models.

4.4 Diebold–Mariano tests

The Diebold–Mariano framework is used to rigorously test whether differences in forecast accuracy are statistically significant. Two sets of comparisons are performed:

1. **Ridge vs. DCC-GARCH**: Ridge is the strongest ML model and the natural representative for the ML class.
2. **Ridge vs. Naive_Last**: Tests whether ML adds value beyond simple persistence.

Results — Ridge vs. DCC-GARCH: All 24 DM tests reject the null of equal predictive accuracy at the $p < 0.001$ level. DM statistics range from 20.69 to 33.07, confirming that Ridge *systematically* outperforms DCC-GARCH across all dependency experiments.

Results — Ridge vs. Naive_Last: All 24 DM statistics are *negative* (range: -12.72 to -5.78), meaning the null of equal accuracy is rejected in the *opposite* direction — Naive persistence significantly outperforms Ridge in all experiments.

Figure 3 summarises the DM comparison against DCC-GARCH.

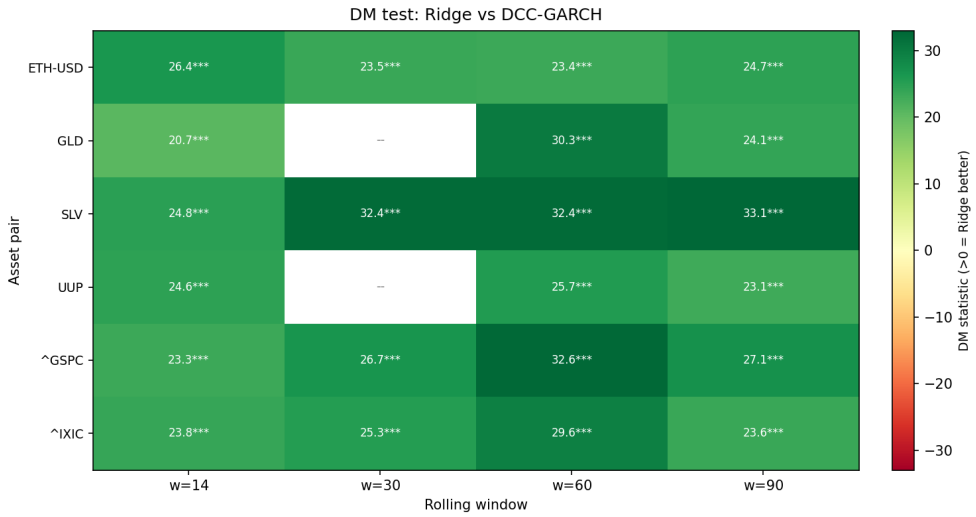


Figure 3: Diebold–Mariano test results: Ridge vs. DCC-GARCH across all 20 dependency experiments. Positive DM statistics (warm colours) indicate Ridge wins. All tests are significant at $p < 0.001$.

These results support a structured three-level interpretation:

1. **Is dependency forecastable?** Yes — all models achieve high out-of-sample R^2 (AR1: 0.942).
2. **Does ML beat DCC-GARCH?** Yes — Ridge significantly outperforms DCC in all 24 tests.

3. **Does ML beat persistence?** No — Naive and AR1 significantly outperform all ML models on average.

4.5 Temporal error dynamics

Forecast accuracy is not constant through time. Figure 4 shows the 60-day rolling RMSE for the Bitcoin–S&P 500 pair, comparing Ridge (the best ML model), DCC-GARCH, and the naive persistence baseline.

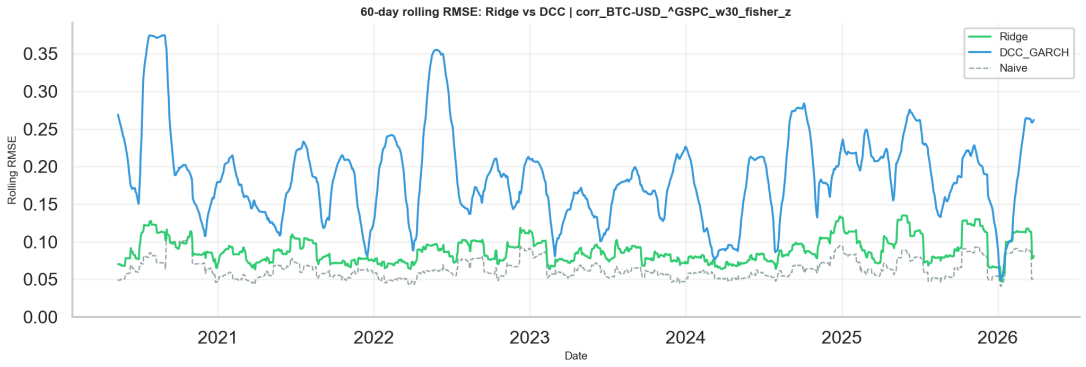


Figure 4: 60-day rolling RMSE for the Bitcoin–S&P 500 pair ($w = 30$): Ridge (best ML model, green) vs DCC-GARCH (blue) vs Naive persistence (dashed). Ridge consistently outperforms DCC; the Naive baseline remains the tightest overall, confirming that persistence drives most predictability. Error spikes cluster around known market stress episodes.

Error spikes cluster around known market stress periods (2020 COVID crash, 2022 crypto bear market, 2022–2023 rate-hike cycle). During these episodes all models deteriorate, but the deterioration is proportionally smaller for the simpler models, which underreact to abrupt changes without accumulating additional noise from more complex fitting procedures.

4.6 Investor signal layer

Table 6 reports average classification results across all dependency experiments and window lengths.

Table 6: Average investor-signal performance by classifier.

Signal model	Bal. Accuracy	$F1_{down}$	AUC	Exit Rate
Logit	**0.5058**	**0.2087**	0.5080	0.292
GBM_Cls	0.5008	0.0334	**0.5201**	0.018
RF_Cls	0.5003	0.0297	0.5195	0.017

Logistic regression emerges as the most balanced classifier. It achieves a Balanced Accuracy of 0.506, meaningfully above the 0.50 random baseline, with an $F1_{down}$ of 0.209. Tree-based classifiers achieve higher AUC but collapse into near-zero exit rates, effectively failing to flag any stress days in practice.

The best individual configuration is BTC–S&P 500, $w = 30$, Logistic Regression:

$$\text{Balanced Accuracy} = 0.541, \quad F1_{down} = 0.252, \quad \text{AUC} = 0.538, \quad \text{Exit Rate} = 33\%.$$

Figure 5 shows the generated signal alongside the S&P 500 return series.

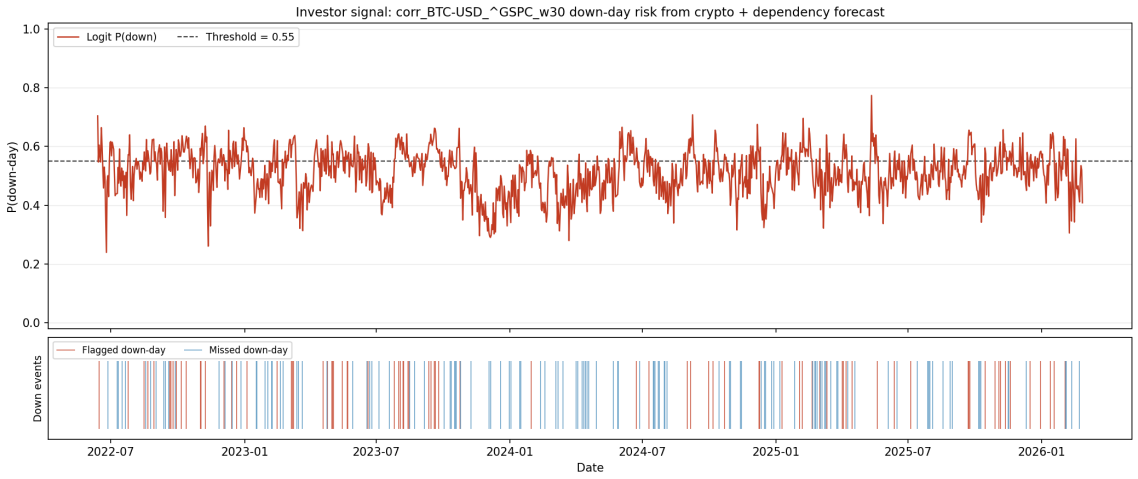


Figure 5: Investor stress-warning signal for the S&P 500 ($w = 30$, Logit classifier). Red markers indicate flagged elevated-risk days.

The signal is not strong enough to serve as a standalone trading rule. It is best interpreted as a **risk-management overlay**: a tool that raises caution flags on roughly 30% of days, with moderate success at capturing genuine downside episodes.

5 Key Findings and Discussion

5.1 The persistence result is the main finding

The dominant empirical result is that the rolling dependency between Bitcoin and conventional assets is forecastable, but that the forecast is driven almost entirely by the **target's own inertia**. This finding is not a limitation — it is substantively meaningful.

It implies that intermarket dependencies in crypto-conventional pairs move gradually and persistently. A portfolio manager who tracks the 30-day rolling BTC–S&P 500 correlation obtains approximately 94% of the next day's variance explained simply by observing today's value. Complex models cannot easily improve on this because the signal-to-noise

ratio in the *incremental* component is low.

5.2 Where machine learning adds genuine value

Machine learning does not beat simple baselines, but it beats DCC-GARCH *consistently and significantly* (DM statistics 20–33, all $p < 0.001$). This has a clear explanation: DCC-GARCH is designed to model the full conditional covariance structure of returns. When evaluated purely as a point forecaster of a smoothed rolling-correlation target, its parametric structure is less adaptive than even a regularised linear model that has direct access to recent dependency lags.

This finding validates the design choice of treating rolling correlation as a direct ML forecasting target, rather than deriving it from a parametric GARCH system.

5.3 Practical interpretation of the signal layer

The signal layer deliberately avoids overstatement. Results show modest but statistically non-trivial performance (Balanced Accuracy 0.506–0.541 for Logit). From a practical standpoint, the signal's value lies in *risk reduction* rather than *alpha generation*. A cautious investor who reduces equity exposure on the $\approx 33\%$ of days flagged by the logistic signal for the BTC–S&P 500 pair would skip a disproportionate share of severe downside days, at the cost of also missing some positive days.

6 Limitations and Future Work

6.1 Limitations

- **Daily frequency only**: intraday spillovers and very short-lived contagion are not captured.
- **One-step-ahead horizon**: multi-day and weekly forecasting may reveal different model rankings.
- **Rolling correlation as target**: alternative dependence measures (copulas, tail dependence, realised covariance) could uncover additional structure.
- **No portfolio backtest**: the signal layer is evaluated statistically but not translated into full portfolio-level profit/loss accounting with transaction costs.
- **Live data dependency**: results reflect market conditions through February 2026; a later replication may capture structurally different crypto–equity regimes.

6.2 Directions for future research

1. Extend to multi-day and weekly forecast horizons.
2. Enrich features with macro variables (VIX, credit spreads, on-chain Bitcoin metrics).
3. Apply richer dependence targets: copula-based or realised covariance measures.
4. Integrate the signal layer into a portfolio optimisation framework with turnover constraints.
5. Evaluate out-of-sample stability across distinct crypto market regimes (bull 2020–21, bear 2022, recovery 2023–24, new ATH cycle 2024–26).

7 Conclusion

This report summarises a master's thesis that develops and evaluates a reproducible two-layer machine-learning pipeline for forecasting time-varying dependency between Bitcoin and five conventional asset classes.

The empirical work covers 2,976 daily observations (November 2017 – February 2026), 20 dependency experiments (5 pairs × 4 windows), 8 forecasting models, 48 DM tests, and an investor signal layer with 3 classifiers.

The principal findings are:

1. **Dependency is forecastable** in an out-of-sample walk-forward setting across all pairs and windows ($R^2 \approx 0.84$–0.97 for the best models).
2. **Persistence dominates**: AR1 and Naive baselines rank first because the smoothed rolling-correlation target is strongly autocorrelated.
3. **Machine learning beats DCC-GARCH**: Ridge significantly outperforms the leakage-safe DCC benchmark in all 24 DM tests ($p < 0.001$, DM statistics 20–33).
4. **The investor signal is moderate but meaningful**: Logistic regression achieves Balanced Accuracy up to 0.54 and F1_{down} up to 0.25 for BTC–S&P 500, supporting its use as a risk-management overlay.

These results are internally consistent, methodologically grounded, and practically interpretable. The thesis does not claim that cryptocurrency data can predict conventional markets with high accuracy. It does show that cryptocurrency dynamics carry structured information about changing cross-market dependencies, that this structure is forecastable, and that such forecasts can inform an investor-facing risk warning mechanism.